

A
PROJECT REPORT
ON
“MINISIM: Security Information and Event Monitoring System”
(using Python and Nmap)

**Submitted in Partial Fulfillment of the requirements for the completion
of the**

**Bachelor of Technology
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DECLARATION

I hereby declare that the **Mini Project** entitled "**MiniSIM: Security Information and Event Monitoring System using Python and Nmap**" is my original work carried out under the guidance of **Mr. Vivekanand Pandey, Assistant Professor, Department of Computer Science & Engineering, Kashi Institute of Technology, Varanasi.**

I further declare that this work has not been submitted, either in part or in full, for the award of any degree, diploma or any other academic qualification at any institution.

Date:

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Signature



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CERTIFICATE

This is to certify that the **Mini Project** entitled "**MiniSIM: Security Information and Event Monitoring System using Python and Nmap**" has been successfully completed by **Sanskriti Srivastava, B.Tech (Computer Science & Engineering), Kashi Institute of Technology, Varanasi**, under my guidance and supervision.

This project fulfills the requirements of the **Mini Project** for the **academic session 2026–2027**.

The work presented in this report is original and has not been submitted elsewhere for the award of any other degree or diploma.

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ABSTRACT

With the rapid growth of digital systems and network-based applications, Cybersecurity threats have become a major concern for organizations. Security Information and Event Management (SIEM) systems help in collecting, analyzing, and monitoring security-related events from different sources to identify potential threats.

MiniSIM (Mini Security Information and Event Management System) is a lightweight SIEM solution developed to demonstrate the basic functionalities of a security monitoring system. The project focuses on collecting system logs, analyzing log data, detecting suspicious activities, and presenting security information through a user-friendly dashboard.

The system is developed using Python and Flask, with SQLite used for database management. It includes modules for log collection, log parsing, threat detection, reporting, and visualization. The threat detection module analyzes collected data and identifies possible security issues based on predefined rules and patterns.

The main objective of this project is to understand the working principles of SIEM systems and implement a simplified security monitoring framework suitable for learning and small-scale environments. This project provides practical knowledge of cybersecurity concepts, log analysis, database handling, and web-based security dashboards.

Keywords: SIEM, Cybersecurity, Log Analysis, Threat Detection, Python, Flask, Security Monitoring.

CHAPTER 2

INTRODUCTION

2.1 Overview

In today's digital era, organizations and individuals rely heavily on computer systems and networks for storing, processing, and exchanging information. With the increasing use of technology, cyber threats such as unauthorized access, malware attacks, and suspicious activities have also increased. Therefore, monitoring and analyzing security events has become an essential part of cybersecurity.

Security Information and Event Management (SIEM) is a cybersecurity solution that collects security-related data from different sources, analyzes it, and helps in detecting potential threats. SIEM systems provide real-time monitoring, event correlation, alert generation, and security reporting.

MiniSIM (Mini Security Information and Event Management System) is a simplified SIEM-based project developed to understand the basic concepts of security monitoring. It collects system logs, processes the collected information, detects suspicious patterns, and displays security-related insights through a dashboard.

2.2 Problem Statement

Traditional systems generate a large amount of log data, making it difficult to manually monitor and identify security issues. There is a need for an automated system that can collect logs, analyze events, and highlight suspicious activities efficiently.

2.3 Objectives of the Project

The main objectives of MiniSIM are:

- To collect and manage system log data.
- To analyze logs for identifying suspicious activities.
- To implement basic threat detection mechanisms.
- To store security events using a database.
- To provide a dashboard for displaying security information.
- To understand the working principles of SIEM systems.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

In traditional security monitoring approaches, system logs are usually checked manually by administrators. This process is time-consuming and may lead to delays in identifying security threats. Large amounts of generated log data make it difficult to analyze events and detect suspicious activities effectively.

Existing enterprise-level SIEM solutions provide advanced security monitoring features, but they are often complex and expensive for beginners and small-scale environments.

3.2 Proposed System

MiniSIM is a lightweight Security Information and Event Management system designed to provide basic security monitoring capabilities. The proposed system collects log data, processes and analyzes events, detects suspicious activities using predefined rules, and displays security information through a web-based dashboard.

The system helps users understand the workflow of a SIEM solution, including log collection, analysis, threat detection, and reporting.

3.3 Requirements Analysis

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: Minimum 5 GB free space
- Operating System: Windows/Linux

Software Requirements

- Programming Language: Python
- Framework: Flask
- Database: SQLite
- Frontend: HTML, CSS, JavaScript
- Development Environment: VS Code
- Libraries: Pandas, NumPy

3.4 Functional Requirements

- Collect system logs from available sources.
- Parse and process log information.
- Detect suspicious activities based on defined rules.
- Store security events in the database.
- Display analysis results through a dashboard.

3.5 Non-Functional Requirements

- User-friendly interface
- Efficient log processing
- Reliable data storage
- Easy maintenance and scalability

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

MiniSIM follows a modular architecture consisting of different components that work together for security monitoring. The system collects log data, processes the collected information, detects possible threats, stores security events, and displays the results through a web-based dashboard.

The major components of MiniSIM are:

1. Log Collection Module

This module collects system log data from available sources and forwards it for further analysis.

2. Log Parsing Module

This module processes raw log data and converts it into a structured format that can be analyzed easily.

3. Threat Detection Module

This module analyzes log events using predefined rules and identifies suspicious activities or security threats.

4. Database Module

The detected events and processed data are stored in the database for future reference and reporting.

5. Dashboard Module

The dashboard provides a visual representation of security events, detected threats, and system information.

4.2 Module Description

Log Collector

The log collector is responsible for gathering system-generated logs and preparing them for analysis.

Log Parser

The parser extracts useful information from raw logs such as timestamps, event details, and severity levels.

Threat Detector

The threat detection component checks events against predefined conditions to identify abnormal or suspicious behavior.

Reporting Module

This module generates reports based on analyzed security events.

Dashboard

The dashboard displays security monitoring results in an understandable format for users.

4.3 Data Flow

The data flow of MiniSIM can be described as:

Log Sources → Log Collector → Log Parser → Threat Detection → Database → Dashboard

The collected logs are processed and analyzed to detect security events. The results are stored and displayed through the dashboard for monitoring.

4.4 System Workflow

1. User starts the MiniSIM application.
2. System collects available log data.
3. Logs are parsed and converted into structured information.
4. Threat detection rules are applied.
5. Security events are stored in the database.
6. Dashboard displays the analysis results.

CHAPTER 5

TECHNOLOGIES AND TOOLS USED

5.1 Programming Language

Python

Python is the primary programming language used for developing MiniSIM. It is used for implementing log processing, threat detection logic, database interaction, and backend functionality due to its simplicity and availability of cybersecurity-related libraries.

5.2 Framework

Flask

Flask is a lightweight Python web framework used to develop the web interface of MiniSIM. It handles backend operations and connects the application logic with the dashboard interface.

5.3 Database

SQLite

SQLite is used for storing security events, processed logs, and detection results. It provides a lightweight and efficient database solution suitable for small-scale applications.

5.4 Frontend Technologies

HTML

HTML is used for creating the structure of the dashboard interface.

CSS

CSS is used for designing and improving the visual appearance of the dashboard.

JavaScript

JavaScript is used for adding interactive features to the user interface.

5.5 Libraries and Modules

Pandas

Used for data handling and analysis of structured log information.

NumPy

Used for numerical operations and efficient data processing.

Flask Libraries

Used for creating routes, handling requests, and connecting the backend with the frontend.

5.6 Security Tools Used

Nmap

Nmap is used for network scanning and identifying available hosts, ports, and services.

5.7 Development Tools

Visual Studio Code

VS Code is used as the development environment for writing, testing, and managing project files.

GitHub

GitHub can be used for version control and maintaining project code.

CHAPTER 6

IMPLEMENTATION

6.1 Overview

The implementation phase focuses on developing the MiniSIM application by integrating different modules for log collection, log analysis, threat detection, data storage, and dashboard visualization. Each module performs a specific function and works together to provide a simple Security Information and Event Management (SIEM) system.

6.2 Log Collection Module

The log collection module gathers system log information from available log sources. These logs are collected continuously and prepared for further processing.

6.3 Log Parsing Module

The collected logs are parsed to extract useful information such as timestamp, event type, severity level, source, and message. The parsed data is converted into a structured format for easy analysis.

6.4 Threat Detection Module

The threat detection module analyzes the processed log data using predefined detection rules. If any suspicious or unusual activity is identified, the system marks it as a potential security threat.

6.5 Database Management

The processed log data and detected security events are stored in the SQLite database. This enables efficient retrieval of information for monitoring and reporting.

6.6 Dashboard

The dashboard provides a graphical interface for displaying collected logs, detected threats, alerts, and other security-related information. It allows users to monitor system activities in an organized manner.

6.7 Report Generation

The system generates reports containing detected security events and analysis results. These reports help users review system activities and understand possible security risks.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 System Execution

The MiniSIM application was successfully developed and executed using Python, Flask, and SQLite. The system was tested in a local environment where logs were collected, analyzed, and processed for security monitoring.

7.2 Log Collection Results

The system successfully collected log data and stored it for further processing. The collected logs were parsed into a structured format, making them easier to analyze.

7.3 Threat Detection Results

The threat detection module analyzed the collected log data and identified suspicious events based on predefined rules. Detected events were stored in the database and displayed on the dashboard for monitoring.

7.4 Dashboard Output

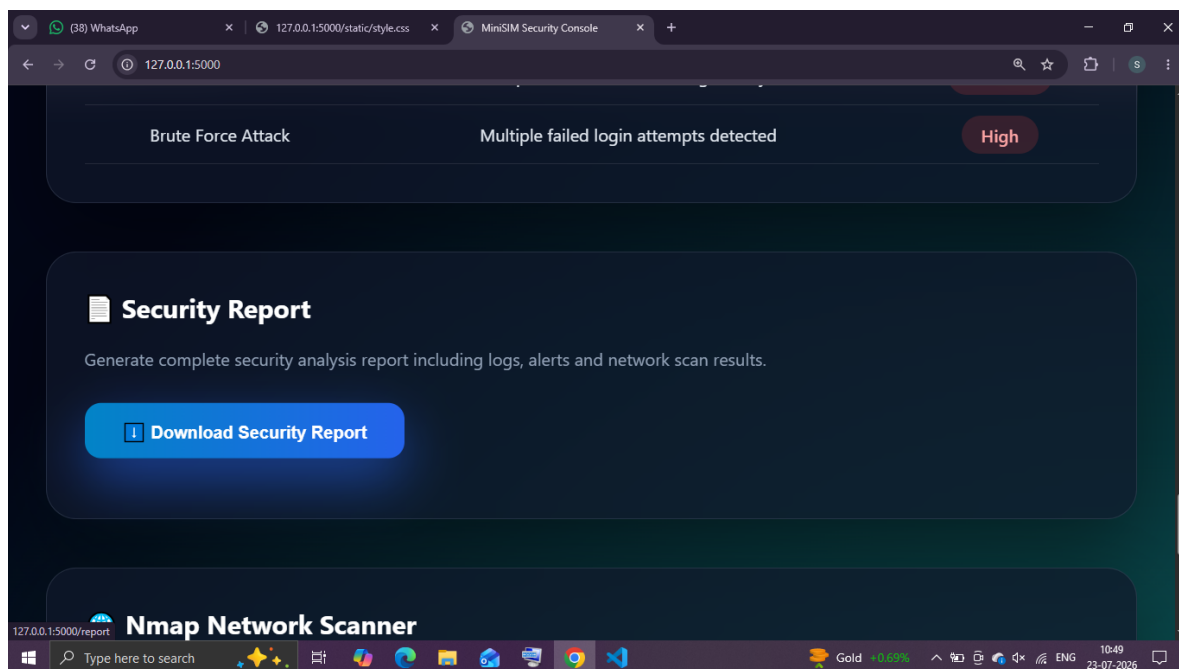
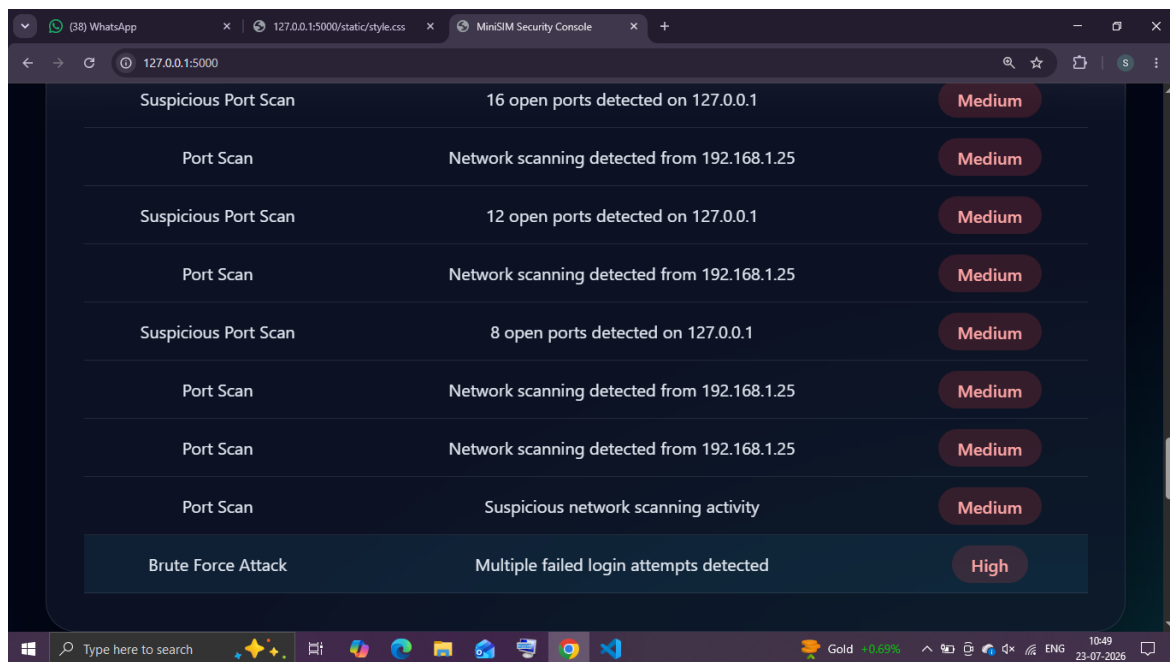
The dashboard displayed important security information such as collected logs, detected threats, alerts, and system status. The graphical interface made it easier to understand the analyzed data.

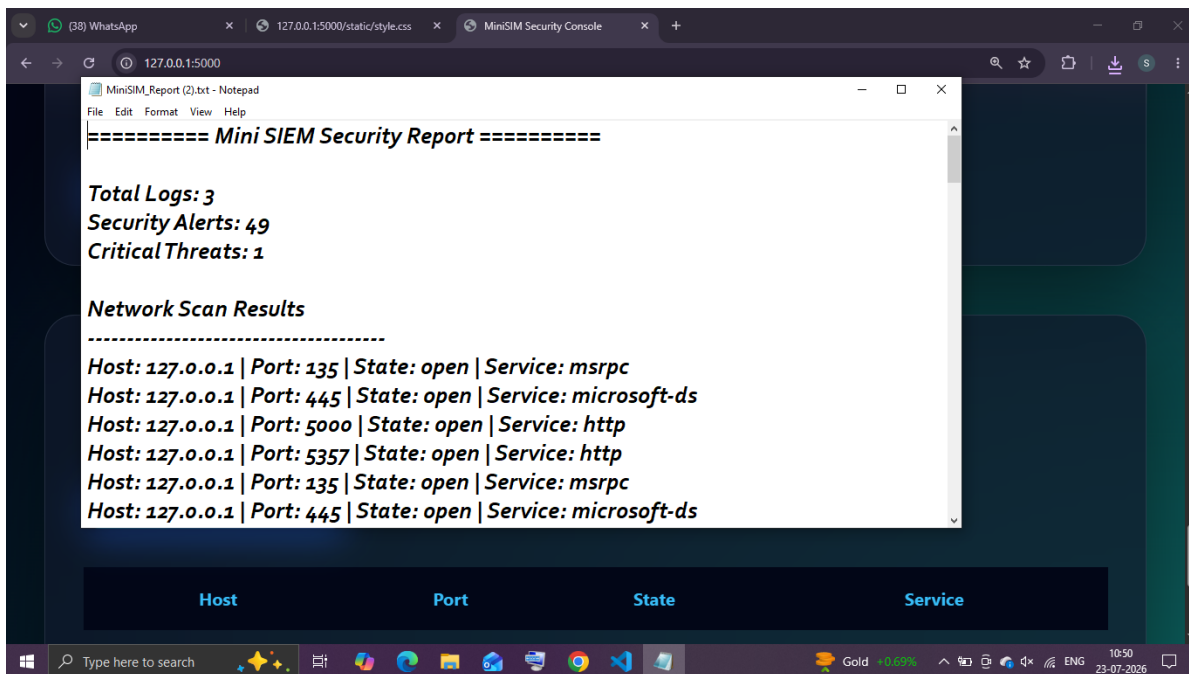
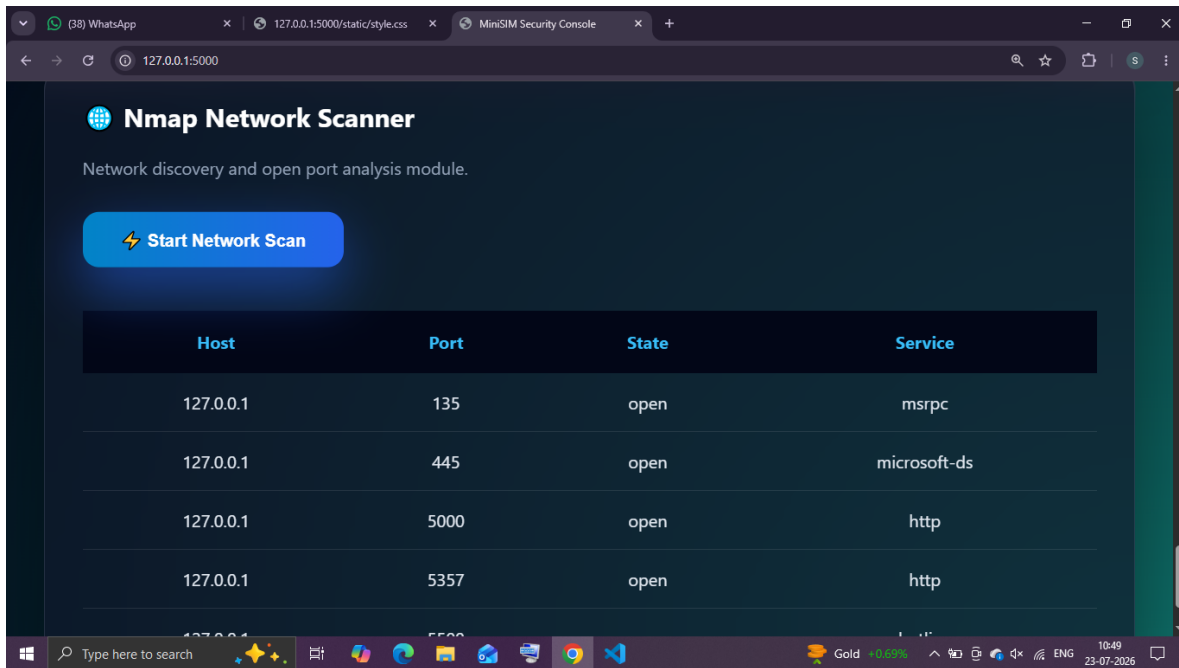


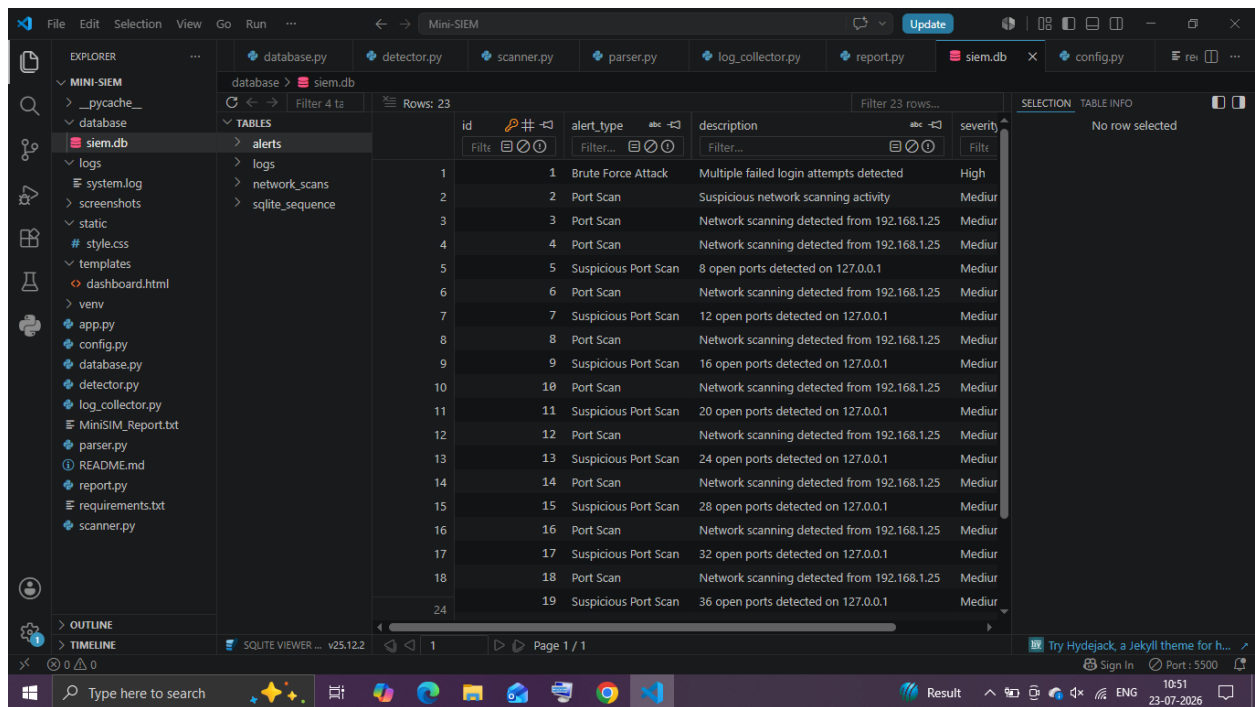
The screenshot shows the "Recent Security Alerts" section of the MiniSIM Security Console. It features a table with three columns: "Alert Type", "Description", and "Severity". The table lists six alerts, all with a "Medium" severity level. The alerts alternate between "Suspicious Port Scan" and "Port Scan" types, with descriptions indicating detected open ports or network scanning on specific IP addresses.

Alert Type	Description	Severity
Suspicious Port Scan	97 open ports detected on 127.0.0.1	Medium
Port Scan	Network scanning detected from 192.168.1.25	Medium
Suspicious Port Scan	92 open ports detected on 127.0.0.1	Medium
Port Scan	Network scanning detected from 192.168.1.25	Medium
Suspicious Port Scan	87 open ports detected on 127.0.0.1	Medium
Port Scan	Network scanning detected from 192.168.1.25	Medium
Suspicious Port Scan	82 open ports detected on 127.0.0.1	Medium

The browser's address bar shows "127.0.0.1:5000". The Windows taskbar at the bottom includes a search bar, application icons, and system tray information showing "Gold +0.69%", "10:48", and "23-07-2026".







7.5 Discussion

The developed MiniSIM system successfully demonstrated the basic working principles of a Security Information and Event Management (SIEM) solution. The project helped in understanding log collection, event analysis, threat detection, database management, and dashboard visualization. Although it is a simplified implementation, it provides a strong foundation for learning cybersecurity monitoring concepts.

CHAPTER 8

CONCLUSION

The MiniSIM (Mini Security Information and Event Management System) project was successfully designed and developed to demonstrate the basic concepts of security monitoring and threat detection. The system is capable of collecting log data, processing security events, detecting suspicious activities based on predefined rules, storing information in a database, and displaying the results through a user-friendly dashboard.

The project provided practical knowledge of cybersecurity concepts such as log management, event analysis, database integration, and web application development using Python and Flask. It also improved the understanding of how SIEM solutions help organizations monitor security events and respond to potential threats.

Although MiniSIM is a simplified implementation, it effectively demonstrates the core functionality of a Security Information and Event Management system and serves as a useful learning platform for students and beginners in cybersecurity.

CHAPTER 9

FUTURE SCOPE

The current version of MiniSIM demonstrates the basic functionality of a Security Information and Event Management (SIEM) system. In the future, the project can be enhanced by adding advanced features to improve its performance and usability.

The following improvements can be considered:

- Integration with multiple log sources such as Windows Event Logs, Linux Syslogs, and network devices.
- Real-time threat detection using machine learning and artificial intelligence techniques.
- Email or SMS alerts for critical security events.
- Interactive dashboards with advanced charts and analytics.
- User authentication and role-based access control for secure access.
- Cloud-based deployment for remote monitoring and scalability.
- Support for automated incident response and threat intelligence integration.

These enhancements would make the system more efficient, scalable, and suitable for real-world cybersecurity monitoring environments.

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APPENDIX

Appendix A: Project Screenshots

- *Figure A.1:* Home Dashboard
- *Figure A.2:* Log Collection Output
- *Figure A.3:* Log Collection Output
- *Figure A.4:* Generate Report Section
- *Figure A.5:* Scan Results
- *Figure A.6:* Generated Report
- *Figure A.7:* Database Records

Appendix B: Project Folder Structure

MiniSIM/

|— *app.py*

|— *config.py*

|— *database.py*

|— *detector.py*

|— *log_collector.py*

|— *parse.py*

|— *report.py*

|— *scanner.py*

|— *requirements.txt*

|— *database/*

|— *log/*

|— *screenshots/*

|— *static/*

|— *templates/*

Appendix C: Software Requirements

- *Python 3.x*
- *Flask*
- *SQLite*
- *VS Code*
- *Windows Operating System*

Appendix D: Hardware Requirements

- *Intel Core i3 or above*
- *Minimum 4 GB RAM*
- *5 GB Free Storage*

GLOSSARY AND ABBREVIATIONS

Glossary

SIEM – A security solution that collects, analyzes, and monitors security events from different sources.

Log – A record of system activities and events generated by devices or applications.

Threat – Any potential event or activity that may compromise the security of a system.

Alert – A notification generated when suspicious or abnormal activity is detected.

Dashboard – A graphical interface used to monitor logs, alerts, and security events.

Database – A structured collection of data used to store logs and security information.

Abbreviations

SIEM – Security Information and Event Management

GUI – Graphical User Interface

DB – Database

SQL – Structured Query Language

HTTP – HyperText Transfer Protocol

HTML – HyperText Markup Language

CSS – Cascading Style Sheets

JS – JavaScript

API – Application Programming Interface